

Pulmonary Mechanics in the Ventilated Patient

High peak-pressure alarm: establish urgency → get a plateau (inspiratory hold) → two subtractions tell you resistance vs compliance. Volume-targeted modes. Adapted from TeachIM.org (McLeod/Scudder; Steinbach; Zhang, Apr 2023), CC BY-NC 4.0.

Key Outputs: set vs monitor

Mode	Monitor (outputs)
Volume-targeted	Peak & plateau (driving) pressures
Pressure-targeted	V_T & minute ventilation V_E

$V_E = V_T \times RR$. Plateau measured on an **inspiratory hold** (no flow).

The two subtractions

Mechanic	Formula	Normal
Resistance	$P_{peak} - P_{plateau}$	<10
Compliance (driving P)	$P_{plateau} - PEEP$	<10*

*for a 4–6 mL/kg breath. Plateau goal **<30**. ARDS driving-pressure goal **≤14–15** (Amato 2015).

Peak = the "everything" pressure (R + C + PEEP + insp flow). Plateau has no flow → resistance drops out → reflects lung/ chest-wall only.

Step 1 — Establish urgency

Emergency = high peak PLUS any of:

- SpO_2 falling
- V_T low / V_E down >50%
- Rising PCO_2
- Dropping BP

→ **Call for help + FiO_2 100%**, work the steps in parallel. Don't delay for an ABG.

Step 2 — Assess resistance

$P_{peak} - P_{plateau} > 10$ = resistance problem.

Cause	Action
Kinked tubing	Inspect / uncompress
Biting ETT	Bite block + sedation/analgesia
Mucus plug / obstruction	Deep suction (± bronch)
Bronchospasm	Bronchodilators
Refractory kink/block	Exchange ETT (airway-trained)

Step 3 — Assess compliance

$P_{plateau} > 30$ or driving > 10 = compliance problem.

Bucket	Causes
Alveolar filling	Edema, ARDS, pneumonia, DAH
Alveolar collapse	Atelectasis, lobar collapse
Extrapulmonary	PTX, pleural effusion, ascites, abdominal distention/ACS, stiff chest wall

Action: history, targeted exam, CXR/POCUS, ABG; bladder pressure if abdominal compartment syndrome suspected.

Tempo & pearls

- **PTX:** spike over min–hrs · **ARDS/fibrosis:** climb over hrs–days · **COPD:** air-trapping (auto-PEEP)
- R and C problems **can co-exist**
- Auto-PEEP / breath-stacking + hypotension → **disconnect circuit** to allow full exhalation, then slow the rate & treat bronchospasm
- Worked Case 1: peak 40, plateau 38, PEEP 12 → $R = 40 - 38 = 2$ (normal); driving = $38 - 12 = 26$ (high) → low compliance (worsening ARDS)

2023–25 framing

- Driving pressure is the vent variable most tied to ARDS survival — keep ≤14–15 even when V_T /plateau already protective (Amato NEJM 2015)
- Lung-protective: V_T 4–6 mL/kg IBW, plateau <30, PEEP >5 (ARDSNet 2000; ATS/ESICM/SCCM 2017; ESICM 2023)
- Global ARDS def 2024: $SpO_2/FiO_2 \leq 315$, lung US, HFNO recognized